EVAN YAP ZHI KAI

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SUTD engineering and entrepreneurship undergraduate who builds and ships AI agents and full-stack products. At Univers, built an audit agent saving ~3 hours per facility audit and an energy model within 2-3 percentage points of observed use; Top 5 of 100 in its website redesign challenge. Seeking AI/software internships; flexible availability.

EDUCATION

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN **Singapore**

Bachelor of Engineering (Engineering Systems and Design), Master of Science in Technology Entrepreneurship *Expected May 2030*

Relevant Coursework: Algorithms & Object-Oriented Programming, Machine Learning, Data Modelling & Visualisation

- **1 of the 6 recipients** of the SUTD Technology Entrepreneurship Programme Scholarship

PROFESSIONAL EXPERIENCE

Univers Pte Ltd | ***Intern - AI Business Consultant (Aviation, Transportation, Logistics - GTM)*** *Aug 2026 - Present*

- Built and deployed an AI Decarbonization & Audit Agent (Next.js, Vercel AI SDK, Groq LPU inference) to automate ESG disclosure ingestion and client claim verification, cutting manual research time by ~3 hours per facility audit.
- Developed and calibrated an HVAC energy-consumption regression model to validate client energy assumptions, achieving estimates within 2-3 percentage points of observed consumption ranges and identifying key efficiency opportunities.
- Analysed operational workflows, industry benchmarks, and stakeholder requirements to define and prioritise AI and energy-management use cases by expected impact, telemetry availability, technical feasibility, and implementation effort.
- Advised the Senior Director, Global Lead, and senior client stakeholders by translating technical analysis into decision-ready business cases, prioritised solution requirements, decarbonisation recommendations, and implementation roadmaps.

EvanYapFitness | ***Founder*** *Sept 2024 - Jan 2026*

- Built an online coaching business to 30 paying clients and $4k in monthly revenue, growing an audience to 2.2k followers and 10k average views per post by testing content and messaging against performance analytics.
- Built a Google Form and Excel intake pipeline and an AI-assisted (Perplexity) workflow that drafts training and meal templates for manual refinement, cutting programming time per client by 50%.

RELEVANT PROJECTS

Univers Website Revamp Challenge (Top 5 of 100 Entries) | JavaScript (ES6+), Three.js (WebGL), CSS3, Vercel

Frontend Developer *Sept 2026*

- Placed ahead of senior solution architects with a 9-page interactive redesign of Univers' EnOS™ Physical AI platform website, and pitched it live to the CEO and senior executives.
- Built it in framework-free modular JavaScript and CSS, with a custom i18n engine serving 7 languages and a real-time Three.js particle mesh that responds to cursor and scroll, lazy-loaded so the site still works without it.
- Shipped an executive ROI modeler for uptime, energy-cost and carbon savings and an edge-to-cloud operations simulator, with reduced-motion fallbacks, accessible markup and automatic Vercel deployment on every push.

Autonomous Trading Agent | Python, Webull OpenAPI, Claude API, SQLite

Developer *Sept 2026*

- Built a Python swing-trading agent on the Webull Singapore API with a deterministic risk engine (position sizing, 1% daily-loss breaker, 5% drawdown limit, automatic stop-loss on fill) and a SHA-256 hash-chained SQLite audit ledger.
- Backtested a momentum baseline over 2 years (Jan 2024 - Sep 2026) at a 1.42 Sharpe and 61% return, then A/B-tested a two-tier Claude (Haiku and Opus) screening layer against it.

Personal Assistant Bot | Next.js, TypeScript, Vercel AI SDK, Gemini, Perplexity Sonar, Google Cloud Pub/Sub, Gmail API

Full-Stack Developer *Aug 2026 - Sept 2026*

- Automated expense capture from Apple Pay (iOS Shortcuts), DBS/PayLah!, Grab and e-commerce receipts via Gmail Pub/Sub push, cutting sync time by ~77% with parallel parsing (Promise.allSettled).
- Designed a 3-tier AI intent engine (Gemini Flash-Lite → 3.6 Flash → Perplexity Sonar failover) routing 24 Zod-validated intents at 100% eval accuracy and ~65% lower cost per call.
- Built a multimodal pipeline that transcribes voice notes and extracts up to 30 events or 25 receipt items per image, including bill splitting, live FX conversion and free-slot finding.
- Hardened security with constant-time auth, prompt-injection guardrails, PII redaction and AES-256-GCM encryption; persisted undo tokens and idempotency logs across serverless cold starts.

KEY SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL, HTML/CSS
Frameworks & Technologies: Next.js, React, Node.js, Vercel AI SDK, REST APIs, Webhooks, Zod, OAuth 2.0, Telegram Bot API
AI & Data: LLM application development, AI agents, prompt engineering, Gemini, Perplexity Sonar, Groq, regression modelling, Excel
Cloud & Developer Tools: Vercel, GitHub, Google Cloud APIs
Leadership: SUTD Men's Floorball Captain, SUTD House Executive Committee (Floor Events & Hostel Works Rep)

ADDITIONAL INFORMATION

Awards: 2nd Place, Vietnam Grand Challenge Hackathon (V-Bây); Best Storytelling nominee (Dragon Tiles)
Interests: Competitive Fitness & Powerlifting (HYROX Athlete), Playing the guitar, Coffee, Volunteer work (Church Worship Ministry)